Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 10: Approved Solutions

Quantifiers and Their Negations

Covers Chartrand & Zhang, Section 3.1

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Ask what the claim would cost

The four jobs

	To show TRUE	To show FALSE
$\forall x \in S, R(x)$	argue for *every* x	one counterexample
$\exists x \in S, R(x)$	one example	argue for *every* x

Decide which box you are in before you write anything. It tells you whether you owe an example or an argument — and half the errors in this section come from owing the wrong one.

Negation in one line

Flip the quantifier, negate the open sentence. With several quantifiers, flip them all, left to right, and negate the open sentence at the end.
And watch the domain. A statement true over $\mathbb{Z}$ can be false over $\mathbb{R}$. Read which one you are given.

Three mistakes to expect

1. **Overshooting.** The negation of "all are" is "at least one is not," never "none are."
2. **Leaving the quantifier alone.** $\sim(\forall x, R(x))$ is not $\forall x, \sim R(x)$.
3. **Swapping quantifier order.** $\forall x \exists y$ and $\exists y \forall x$ are different statements.

Part II: Practice Problems

Section A — True or false

1. Determine the truth value of each, with a one-line justification.

(a) $\forall n \in \mathbb{Z},\ n^2 \geq n$.

(b) $\exists n \in \mathbb{Z},\ n^2 = 2$.

(c) $\forall x \in \mathbb{R},\ x^3 = x$.

(d) $\exists x \in \mathbb{R},\ x^3 = x$.

(e) $\forall n \in \mathbb{Z},\ 3n + 1$ is even.

Approved solution

(a) True. For $n \geq 1$, multiplying $n \geq 1$ by n gives $n^2 \geq n$; for $n \leq 0$, $n^2 \geq 0 \geq n$. The near miss is $n = 1$, where the two are equal — and $\geq$ permits that.
(b) False. No integer squares to 2; $1^2 = 1$ and $2^2 = 4$, and squares grow from there. Showing an existential false requires ruling out the whole domain, not testing a few.
(c) False. Counterexample $x = 2$: $8 \neq 2$. One counterexample suffices.
(d) True. $x = 0$ works. So do 1 and -1. One example suffices.
(e) False. Counterexample $n = 1$: $3(1) + 1 = 4$ is even — wait, that makes the claim true at $n = 1$. Take $n = 2$ instead: $3(2) + 1 = 7$ is odd. One counterexample is enough.

Compare (c) and (d). Same open sentence, different quantifier, opposite truth values. Part (c) took one counterexample; part (d) took one example. Parts (a) and (b) each took a general argument. Which box you are in decides everything.

Section B — Negation

2. Write the negation of each, phrased the way a person would say it.

(a) Every student in this class has a laptop.

(b) Some prime number is even.

(c) $\forall n \in \mathbb{Z},\ 5n + 2$ is odd.

(d) $\exists x \in \mathbb{R},\ x^2 + x + 1 = 0$.

(e) There is no smallest positive real number.

Approved solution

(a) At least one student in this class does not have a laptop. *Not* "no student has a laptop."
(b) No prime number is even — equivalently, every prime is odd. (The original is true, witnessed by 2, so this negation is false.)
(c) There is an integer n for which $5n + 2$ is even.
(d) For every real x, $x^2 + x + 1 \neq 0$. (True — the discriminant is $1 - 4 = -3 < 0$ — so the original is false.)
(e) There is a smallest positive real number. Note the original already contains a negation; negating it *removes* one rather than adding another.

Part (a) is the diagnostic. "No student has a laptop" is the overshoot, and it is the single most

common error in this section.

Section C — Two quantifiers

3. For each, state whether it is true or false over $\mathbb{Z}$, and justify.

(a) $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z},\ ab = a$

(b) $\exists b \in \mathbb{Z}, \forall a \in \mathbb{Z},\ ab = a$

(c) $\exists a \in \mathbb{Z}, \forall b \in \mathbb{Z},\ ab = 0$

(d) $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z},\ a + b = 1$

Approved solution

(a) True. Given a, take $b = 1$.
(b) True. Take $b = 1$, which works for *every* a. Both orders happen to hold here.
(c) True. Take $a = 0$; then $ab = 0$ for every b.
(d) True. Given a, take $b = 1 - a$.

All four are true — deliberately. It is tempting to learn a shortcut like "$\exists\forall$ is usually false." Parts (b) and (c) show that shortcut failing. Order matters, but it does not always flip the answer, and each case must be checked on its own.
Contrast with the pair in Lecture 10, where $\forall x \exists y,\ x + y = 0$ was true and $\exists y \forall x,\ x + y = 0$ was false. That is the shape students remember; these four are the corrective.

4. Write the negation of each in symbols, then in words.

(a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ y^2 = x$

(b) $\exists a \in \mathbb{Z}, \forall b \in \mathbb{Z},\ ab \geq 0$

Approved solution

(a) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R},\ y^2 \neq x$.
In words: there is a real number that is not the square of any real number. *True* — take $x = -1$ — which shows the original statement is false.
(b) $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z},\ ab < 0$.
In words: for every integer a there is an integer b with ab negative. *False* — $a = 0$ gives $ab = 0$ for every b. So the original is true, witnessed by $a = 0$.

Both quantifiers flip and the open sentence is negated. Note $\geq$ becomes $<$, not $\leq$ — the same inequality slip from Week 1, and still the most expensive small error available.

Section D — Writing

5. In five to eight complete sentences, explain why disproving a statement beginning $\forall$ requires only one counterexample, while disproving one beginning $\exists$ requires an argument covering the whole domain. Illustrate with two specific statements of your own.

Approved solution

Model answer. A universal statement claims an open sentence holds at every element of the domain, so a single element where it fails makes the whole claim false — the negation of $\forall x, R(x)$ is $\exists x, \sim R(x)$, and producing that one x is exactly what a counterexample does. For instance, "every prime number is odd" is disproved instantly by 2, and no further work is needed. An existential statement claims only that *some* element works, so refuting it means ruling out every element at once; the negation of $\exists x, R(x)$ is $\forall x, \sim R(x)$, which is itself a universal claim. To disprove "there is a real number x with $x^2 + 1 = 0$" you cannot test values — you argue that $x^2 \geq 0$ for all real x, so $x^2 + 1$ is never zero. The two jobs differ because negation converts each quantifier into the other, and the universal side is always the expensive one.

Full credit requires the connection to the negation rules, not merely the observation that one takes an example and the other an argument.

Section E — Challenge (optional)

6. A statement of the form $\forall x \in S,\ P(x) \Rightarrow Q(x)$ can be false in only one way. Describe it precisely, say what a counterexample must therefore look like, and give a counterexample to: *for every integer n, if n^2 is even then n is a multiple of 4.*

Approved solution

The statement is false exactly when some $x \in S$ makes $P(x) \Rightarrow Q(x)$ false, and an implication is false only when the hypothesis is true and the conclusion is false. So a counterexample must be a single $x \in S$ with $P(x)$ **true** and $Q(x)$ **false**. Both halves are required.
Counterexample: $n = 2$. Then $n^2 = 4$ is even, so the hypothesis holds; but 2 is not a multiple of 4, so the conclusion fails. ($n = 6$ works too.)

This is the shape of nearly every disproof you will write from Chapter 3 onward. Students who offer an x failing the *hypothesis* — here, an odd n — have found a row where the implication is *true*, which supports the claim rather than refuting it.

Instructor notes

- **Problem 2(a)** is the overshoot diagnostic. If a third of the class writes "no student has a laptop," spend five minutes on it before Homework 5.
- **Problem 3 is the corrective to Lecture 10.** All four are true, which breaks the shortcut "$\exists\forall$ is usually false." Discuss alongside the lecture pair where order *did* flip the answer.
- **Problem 6 is the bridge to Chapter 3.** Every disproof students write from Week 5 onward has this shape: hypothesis true, conclusion false. Naming it now saves time later.
- **Problem 1(e)** rewards care — $n = 1$ is not a counterexample, $n = 2$ is. Students who grab the first value they try will get it wrong.
- Reserved for Homework 5: Section 3.1 Exercises 12, 14, 16, 18 (even). Odd-numbered exercises have printed answers at pp. 596–597 and are self-check only.